Glycogen: Glycogen is the storage polysaccharide of animals. It is present in the liver and muscles of animals. Glycogen is also called as animal starch. On hydrolysis it gives glucose molecules. Structurally, glycogen resembles amylopectin with more branching. In glycogen the branching occurs every 8-14 glucose units opposed to 24-30 units in amylopectin. The excessive glucose in the body is stored in the form of glycogen.

14.1.7 Importance of carbohydrates

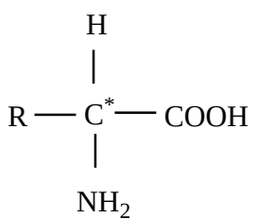
1. Carbohydrates, widely distributed in plants and animals, act mainly as energy sources and structural polymers.
2. Carbohydrate is stored in the body as glycogen and in plant as starch.
3. Carbohydrates such as cellulose which is the primary components of plant cell wall, is used to make paper, furniture (wood) and cloths (cotton)
4. Simple sugar glucose serves as an instant source of energy.
5. Ribose sugars are one of the components of nucleic acids.
6. Modified carbohydrates such as hyaluronate (glycosaminoglycans) act as shock absorber and lubricant.

14.2 Proteins

Proteins are most abundant biomolecules in all living organisms. The term protein is derived from Greek word '*Proteious*' meaning primary or holding first place. They are main functional units for the living things. They are involved in every function of the cell including respiration. Proteins are polymers of α-amino acids.

14.2.1 Amino acids

Amino acids are compounds which contain an amino group and a carboxylic acid group. The protein molecules are made up α-amino acids which can be represented by the following general formula.



There are 20 α-amino acids commonly found in the protein molecules. Each amino acid is given a trivial name, a three letter code and a one letter code. In writing the amino acid sequence of a protein, generally either one letter or three letter codes are used.

14.2.2 Classification of α-amino acids

The amino acids are classified based on the nature of their R groups commonly known as side chain. They can be classified as acidic, basic and neutral amino acids. They can also be classified as polar and non-polar (hydrophobic) amino acids.

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